

 Multi-Agent Autonomous Intelligence

MARKETMIND AI

AI-Powered Multi-Agent Market Research & Competitive Intelligence Platform

 BS Computer Science — 8th Semester

 Agentic Workflow System

Executive Overview



What is MarketMind AI?

MarketMind AI is an advanced, autonomous market research intelligence system engineered to transform raw, natural-language research prompts into deep, structured research deliverables.

By orchestrating a collaborative swarm of specialized AI agents, it dynamically plans, searches, verifies, compares, and synthesizes complex market domain insights without human intervention.



Core Value Proposition

Eliminates fragmented manual intelligence gathering by automating search, evidence extraction, dynamic metric calculation, and factual verification into a single cohesive pipeline.

Key Highlight: Guarantees output precision using Pydantic data validation schemas and dedicated Quality Control auditing agents.

Problem Statement



Time-Consuming

Traditional manual market research requires dozens of hours spent gathering fragmented information across inconsistent internet sources.



Scattered Sources

Data remains isolated across earnings reports, web articles, financial filings, and market statistics without a unified intelligence layer.



Inconsistent Quality

Manual analytical methods yield variable depth, unsupported conclusions, and potential confirmation bias in strategic report generation.



Single-Agent Limits

Standard single-prompt LLMs struggle with complex multi-step research, lacking tool access, self-reflection, and structured error handling.



The Solution: Agentic Automation

There is a critical need for an automated, multi-agent orchestration architecture that guarantees structured evidence verification, metric transparency, and repeatable quality control.

Key Objectives



1. Understand Requests

Parse natural-language user prompts into explicit research parameters and constraints.



2. Autonomous Planning

Formulate granular research plans and breakdown queries systematically.



3. Gather Evidence

Query web & internal tool registries to aggregate verifiable evidence records.



4. Entity Comparison

Perform structured comparative matrices for competitors and market players.



5. Synthesize Insights

Identify overarching trends, growth vectors, and risk profiles in target domains.



6. Quality Control

Validate claims against evidence records to purge hallucination and gaps.



7. End-to-End Report Generation

Produce standard executive-ready market research documents with complete source transparency and interactive visualizations.

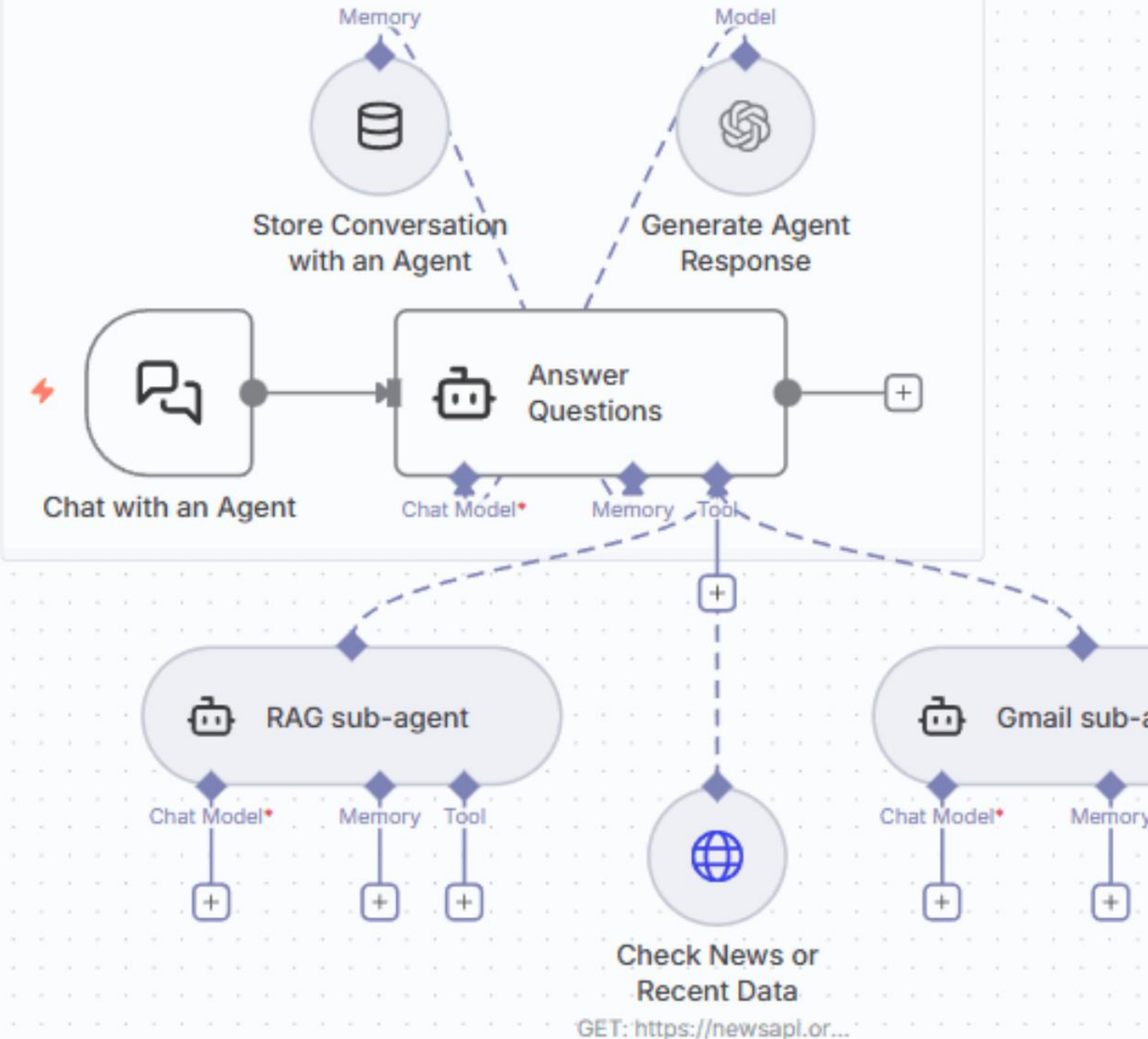
Overall Pipeline Architecture

MarketMind AI operates as a sequential and reflective multi-agent execution pipeline. Each specialized agent transforms input state and passes structured Pydantic objects downstream.

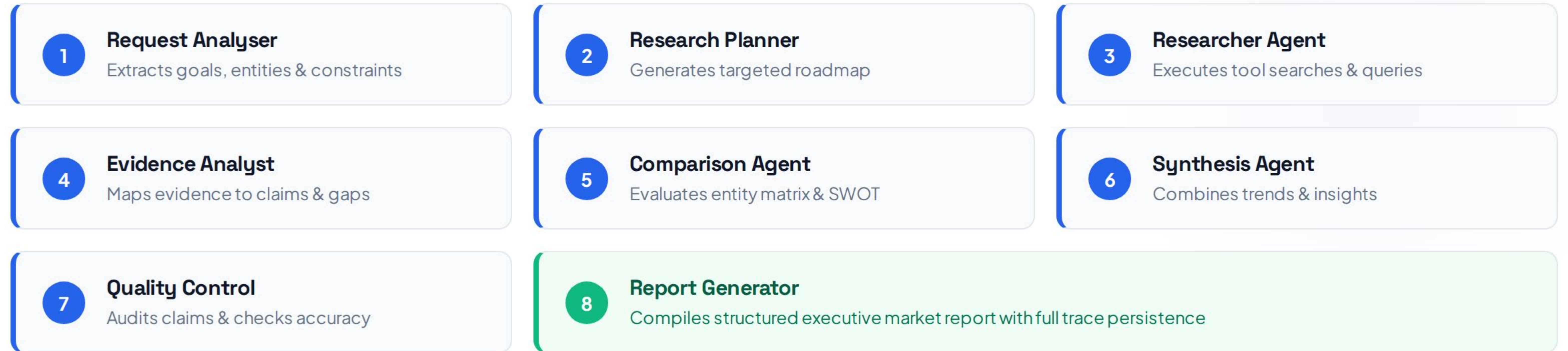
- ▶ **User Request:** Natural language research target
- 🔧 **Multi-Agent Swarm:** 8 Specialized agents executing tools
- 🔍 **Quality Gatekeeper:** Strict validation before final output
- 📄 **Final Report:** Structured market intelligence output

Main AI Agent

Routes user queries to the corresponding Tools (news tool) and subagents (RAG and Gmail sub-agents).



Agent Execution Workflow



■ Core Agents (1/2)

Request Analyser

Parses raw input into structured JSON payload. Extracts key variables:

-  Research Objectives & Industry Domain
-  Target Competitor Entities & Geographies
-  Timeframes, Required Metrics & Constraints

Research Planner

Decomposes macro objectives into micro actionable execution plans:

-  Formulates specific research questions
-  Maps required tools & metric formulas
-  Establishes sequential query priority

■ Core Agents (2/2)



Evidence Analyst

Filters raw data, attaches citations to research questions, highlights supported claims, and tags missing evidence gaps.



Comparison Agent

Evaluates competitive strengths, weaknesses, product features, technologies, and market metrics in tabular format.



Synthesis & QC

Synthesizes narrative trends while Quality Control audits for hallucinations, contradictions, and missing parameters.

Tool Registry & Dispatcher

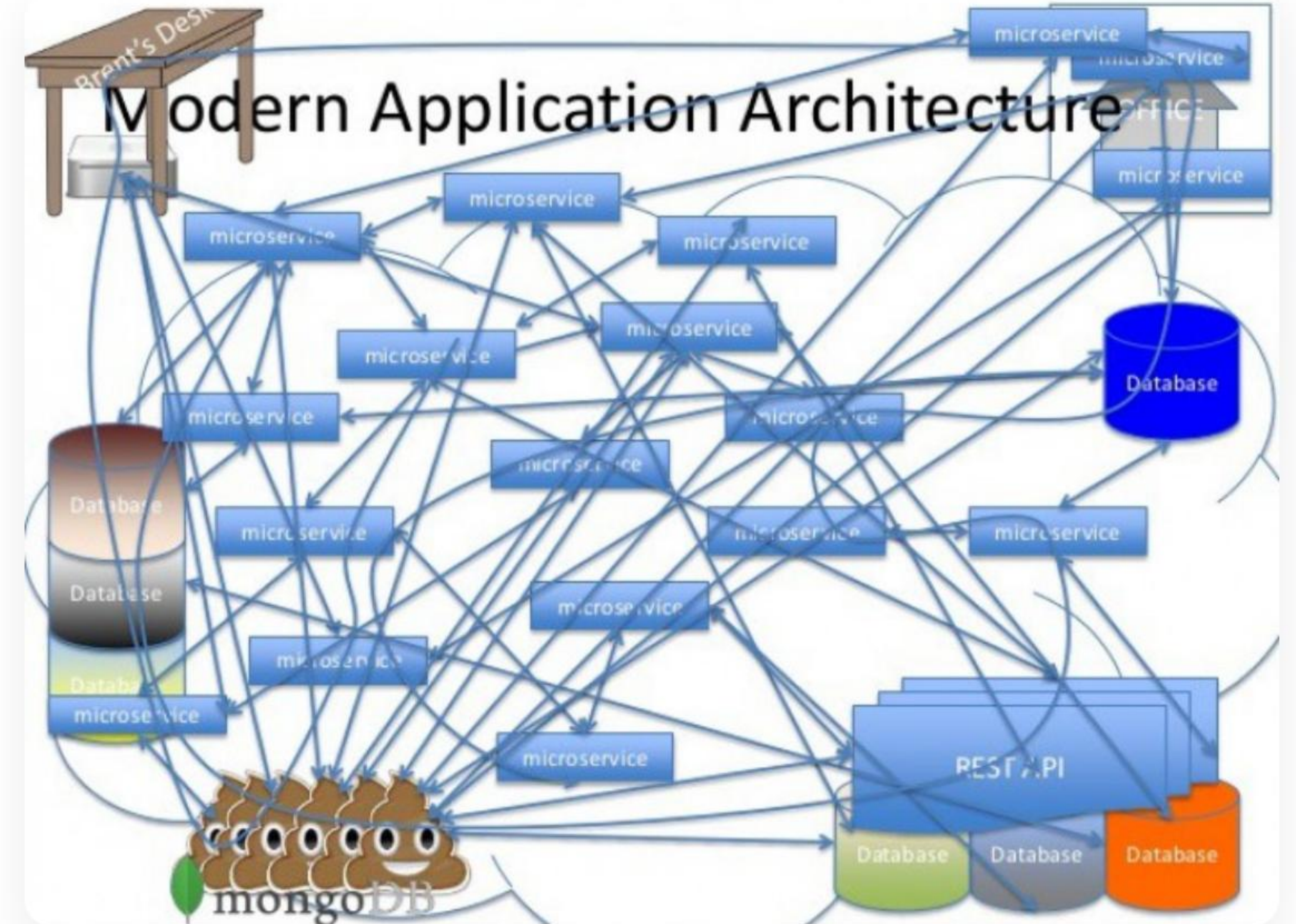
MarketMind AI features an decoupled Tool System that isolates function execution from LLM reasoning.

Tool Registry

Maintains tool metadata, definitions, required arguments, and execution permissions.

Tool Dispatcher

Validates LLM tool arguments, dispatches function execution, captures exceptions, and returns formatted observations.



Reusable Tool Ecosystem



search_information

Retrieves real-time market data, company news, and web intelligence.



retrieve_document

Fetches specific internal corpus files, PDF filings, and saved data.



calculate_metric

Executes deterministic financial calculations and market math.



compare_companies

Generates normalized competitive benchmarking matrices.



save_research

Persists intermediate observations into state storage.



generate_report

Compiles final markdown and JSON structured report outputs.

Pydantic Data Schemas

Schema Name	Core Responsibility	Key Advantages
RequestAnalysis	Extracts objectives, entities, timeline, questions	Guarantees complete input structure
ResearchPlan	Contains research sub-questions & tool mappings	Enforces structured execution paths
EvidenceRecord	Tracks source URI, claim verification & gap tag	Ensures 100% traceably backed facts
MarketReport	Executive summary, findings, SWOT, risks, conclusion	Standardized downstream consumption

Data Integrity Guarantee: Type safety prevents runtime parsing errors, enforcing zero hallucinated object formats across agent boundaries.

■ LLM Integration & Prompting

🧠 Centralized LLM Client

- 🔌 Managed interface wrapper for OpenAI API integration
- 🔄 Automatic retry loops for transient API failures
- ⚙️ Configurable temperature, token limits, and model tiers

>_ Dedicated System Prompts

- 👤 Role definition & explicit operational boundaries
- 🚫 Anti-hallucination rules & zero-assumption constraints
- </> Structured JSON / Pydantic target response formatting

Streamlit User Interface

The interactive Streamlit dashboard enables seamless human-in-the-loop market intelligence exploration.

- Secure UI API key authentication
- Real-time agent progress tracking
- Live Quality Control audit feedback
- Single-click JSON & Markdown exports

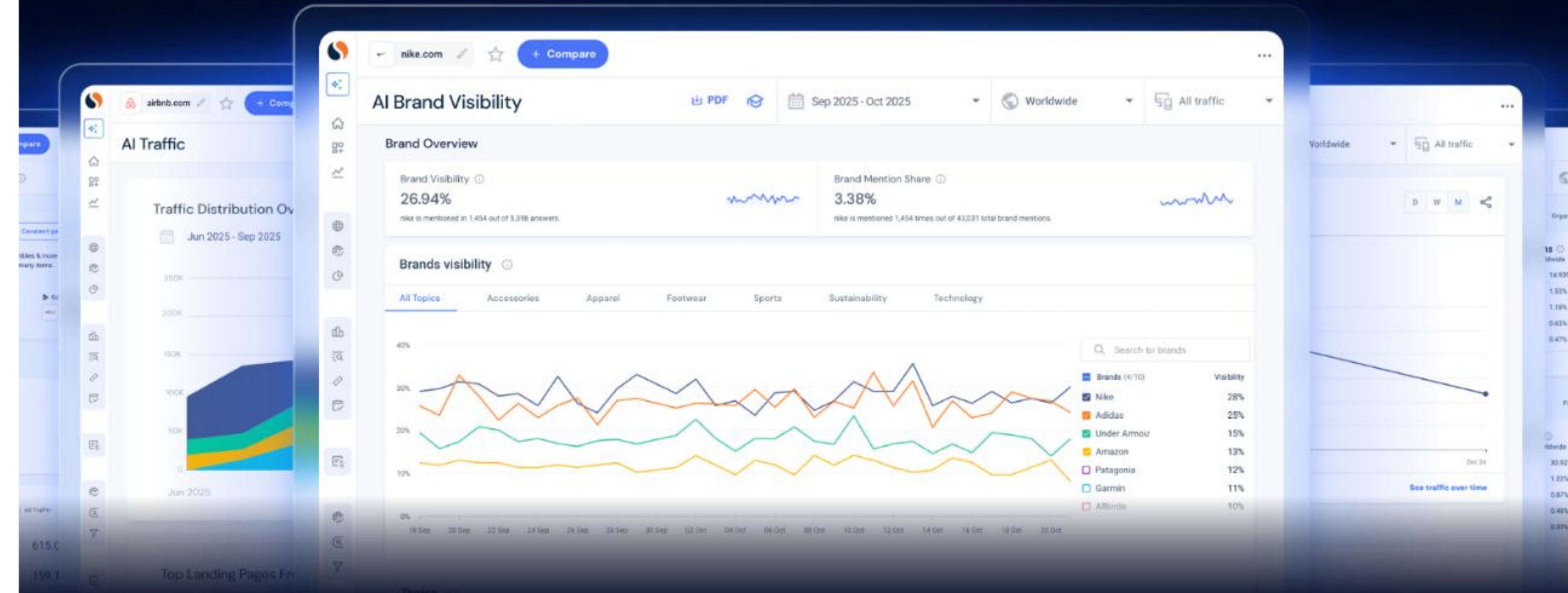
Win Your Market with AI-Powered Digital Data

All your digital channels. One holistic view.

Website

Enter a domain

Get insight



Trusted by the Best and Brightest Digital Brands

ebay

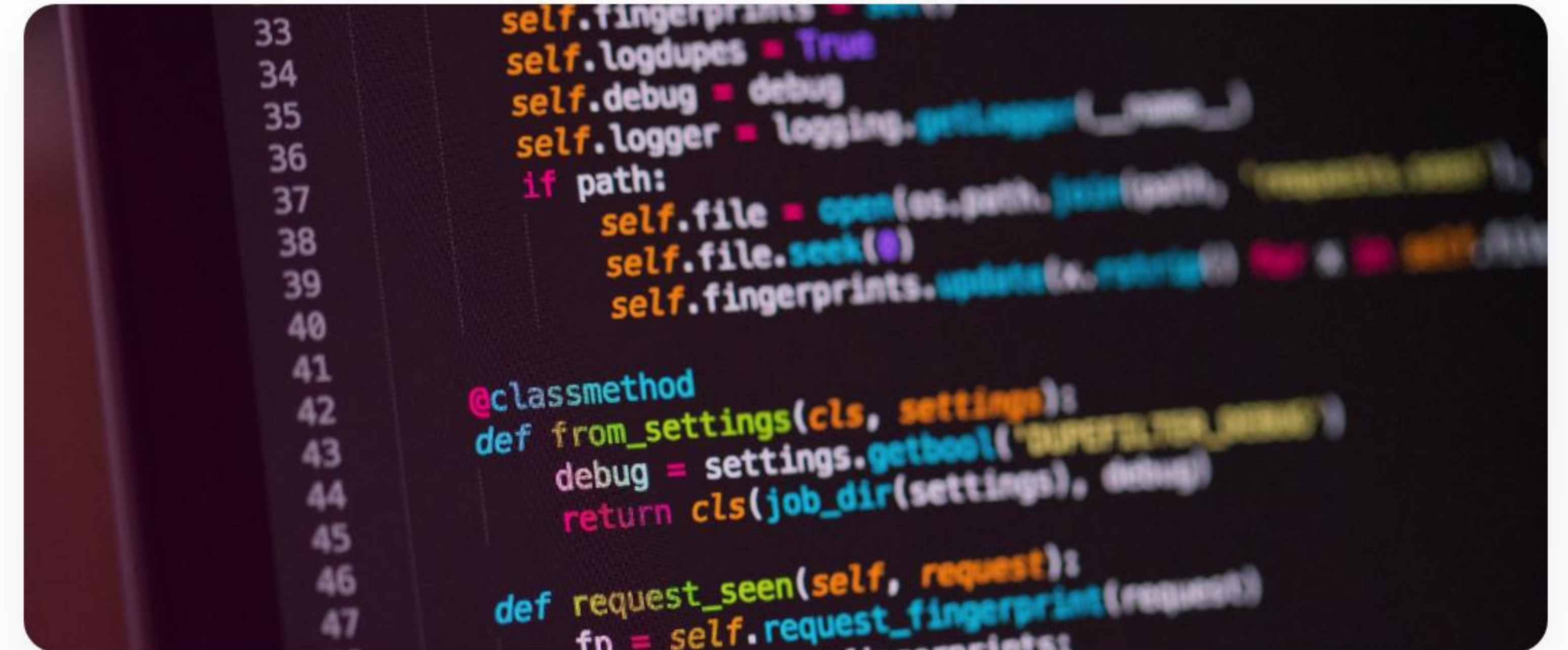
The Economist

DHL

Omnicom

Project Structure & Tech Stack

```
marketmind-ai/  
├── app.py # Streamlit Application Entry  
├── requirements.txt  
├── src/  
│   ├── agents/ # 8 Specialized Agents  
│   ├── schemas/ # Pydantic Schemas  
│   ├── tools/ # Registry & Dispatcher  
│   ├── state/ # State Management  
│   └── llm/ # Client & Prompts  
└── runs/ # JSON Run History Persistence
```



```
33 self.fingerprints = ...  
34 self.logdups = True  
35 self.debug = debug  
36 self.logger = logging.getLogger(__name__)  
37 if path:  
38     self.file = open(os.path.join(path, 'requirements.txt'),  
39                     'r')  
40     self.file.seek(0)  
41     self.fingerprints.update(self.file.read())  
42  
43 @classmethod  
44 def from_settings(cls, settings):  
45     debug = settings.getbool('debug', False)  
46     return cls(job_dir(settings), debug)  
47  
48 def request_seen(self, request):  
49     fn = self.request_fingerprint(request)
```

Python 3.10+

OpenAI API

LangChain

Pydantic

Streamlit

Pytest

■ Testing & Security Standard

Automated Pytest Suite

Comprehensive unit & integration tests validate entire agent workflow integrity:

```
python -m pytest -v
```

- ✓ Schema validation & tool dispatcher tests
- ✓ Agent loop & evidence gap detection tests

Security & Resilience

-  Zero hardcoded secrets; strict .env variable usage
-  Masked API key inputs in UI display
-  Tool exception boundaries managed by dispatcher

Future Enhancements



Live Web Scraping

Integrate real-time SERP APIs and dynamic site scraping agents.



RAG / Vector DB

Embed ChromaDB / Pinecone for multi-gigabyte corpus search.



Financial Connectors

Direct API integration with SEC EDGAR, Bloomberg & Yahoo Finance.



Cloud Multi-Tenant

Deploy scalable multi-user cloud workspace on AWS / GCP.



PDF Export Engine

Automated pixel-perfect PDF/DOCX report formatting engine.



Citation Verification

Automated fact-checking cross-validation models for sources.

✔ MarketMind AI Complete

THANK YOU!

Questions & Strategic Discussion

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Multi-Agent AI Platform

Image Sources



<https://storage.ghost.io/c/0d/78/0d78b34c-0c5f-4975-900e-61d00ccb1c2d/content/images/2025/12/MAS10.png>

Source: blog.n8n.io



<https://sufficiently-advanced.technology/assets/images/web-architecture.jpg>

Source: sufficiently-advanced.technology



https://cdn.prod.website-files.com/668f1109d931de5cd16d38c0/69ee8cac79459e6a0f2ee720_Similarweb.png

Source: www.simular.ai



https://assets.qlik.com/image/upload/w_1792/q_auto/qlik/glossary/data-analytics/seo-data-science-vs-data-analytics-model-development_zcw1xm.jpg

Source: www.qlik.com